Estimation of cycle average convective heat transfer coefficient on the underside of a racing piston for different layouts of oil jets

Gaspare Argento

Outline

- The main goal of this work was to find an alternative nozzle layout having a lower oil flow rate but with the same heat transfer performance. The benefits of having less oil inside the engine are, in general, many.

- Three geometries, different from the reference one for number of holes, diameters and trajectories, have been studied. Here the details:
  - Reference geometry: one oil jet of 1 mm diameter;
  - Proposal P1: two oil jets with a diameter of 0.5 mm;
  - Proposal P2: as P1 in terms of trajectories, but with two holes with a diameter of 0.7 mm;
  - Proposal P3: two oil jets with a diameter of 0.6 mm and slightly different trajectories with respect to proposals P1 and P2.

- All simulations were performed using the commercial mesh-free CFD software Particle Works. The low computational cost of this approach allowed the analysis of five engine speeds, from 8000 to 16000 rpm with a pace of 2000 rpm.

Outline

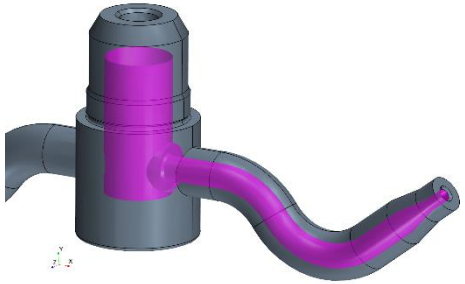
- For computational reasons, the air motion contribution and its influence on the oil jets were not considered, similarly the oil flowing from the conrod and the main bearings was not taken into account.

- The comparison of the heat transfer performances of the tested configurations were made on the basis of the cycle average heat transfer coefficient maps at the underside surface of the piston. Surface averaged heat transfer values were also carried out in order to have a more quantitative measure.

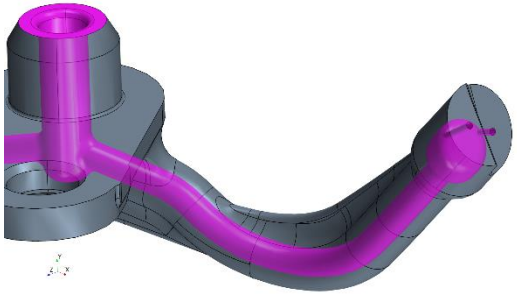
Layout Comparison

- The figures below show in detail the reference geometry and the proposals layout.

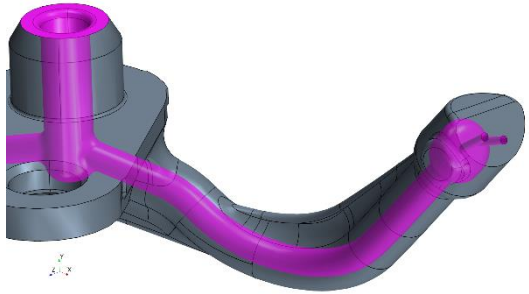
- In all cases the oil jets impact the underside surface of the piston in the central channel, mainly at the intake side. All the other hollows are not directly lubricated, so the heat transfer on these surfaces will be essentially zero.



REFERENCE (1X1.0)



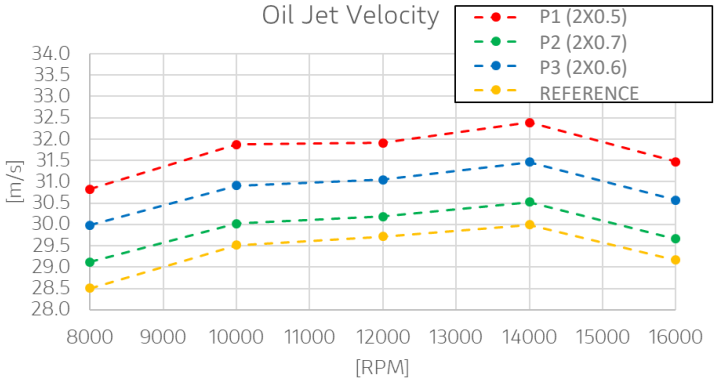
P1 & P2 (2X0.5 & 2X0.7)

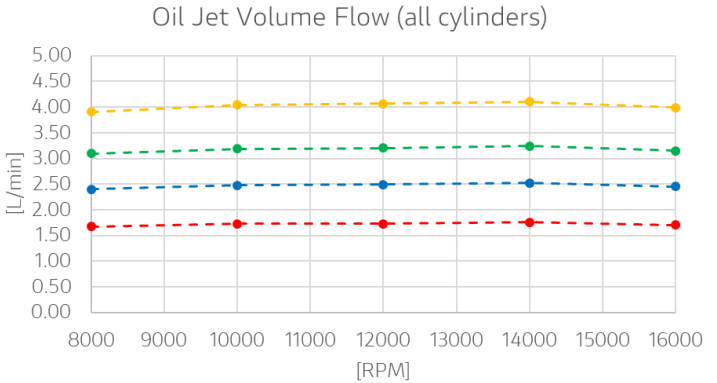


P3 (2X0.6)

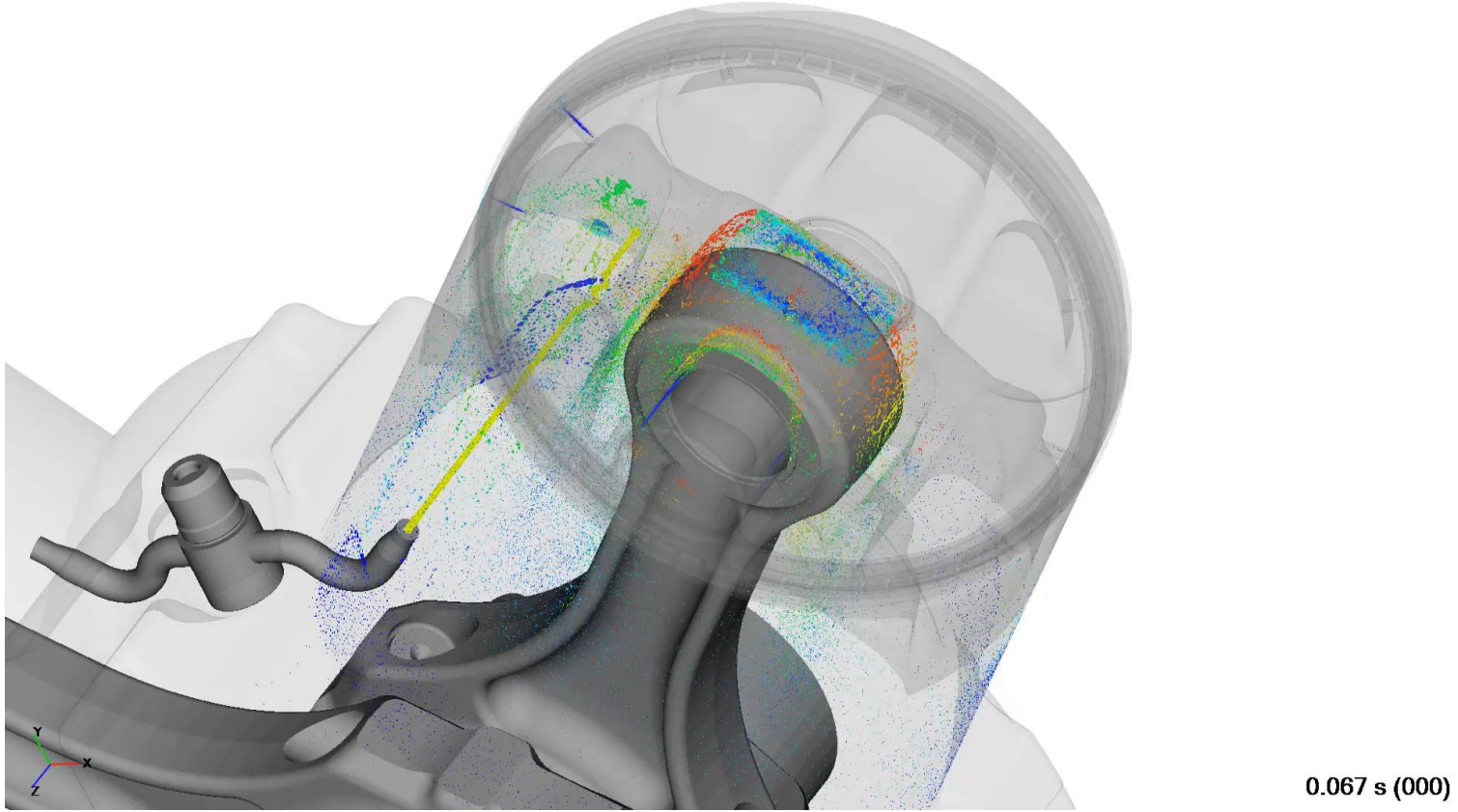
Oil Jet Boundary Conditions

- Oil jet velocities are available from the 1D model of the oil circuit for each layout and are shown in the top graph. The two 0.5 mm jets of proposal P1 are the fastest while the reference one is the slowest. Proposals P2 and P3 velocities fall somewhere in between.
- In terms of volume flow rate values, the scenario is totally reversed as illustrated in the bottom graph: thanks to a bigger area and a higher discharge coefficient, the reference geometry ends up having the highest flow rate. Proposal P1 does not reach its half (the surface area of the two 0.5 mm holes is 50% less than the one of the single 1 mm jet).
- Therefore the question is: will the higher speeds compensate the lower flow rates on the overall heat transfer performance?

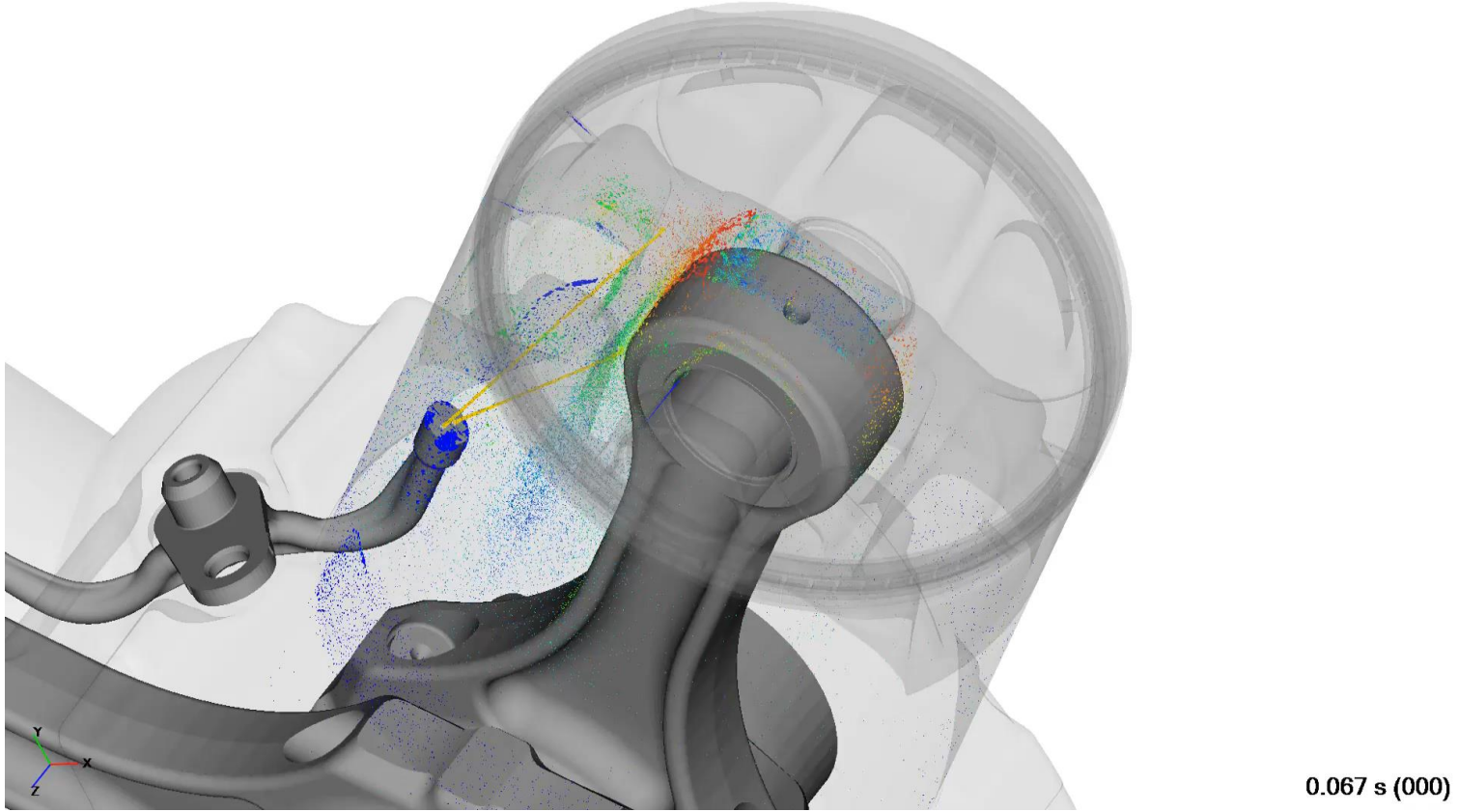




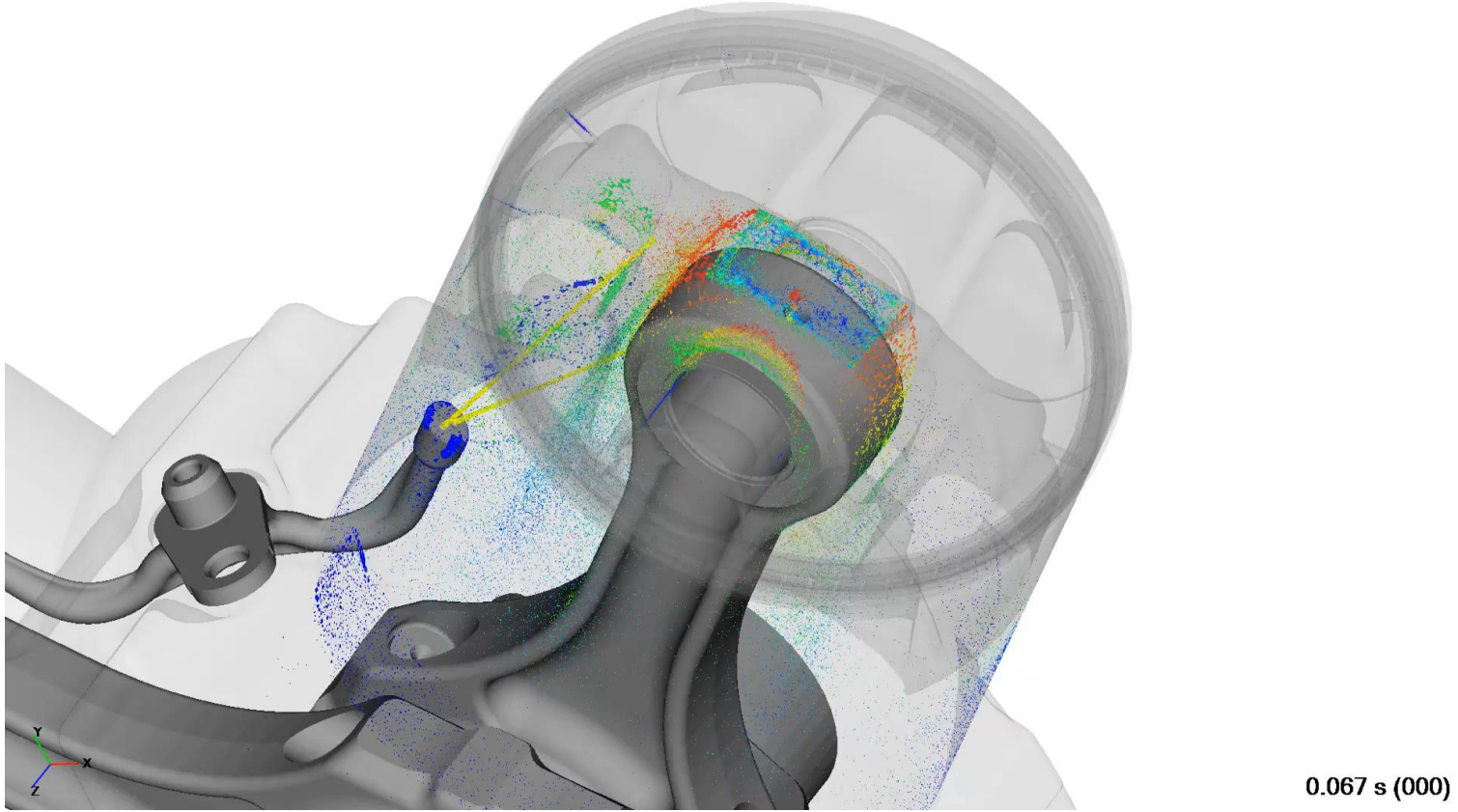
Results: reference geometry (1X1.0)



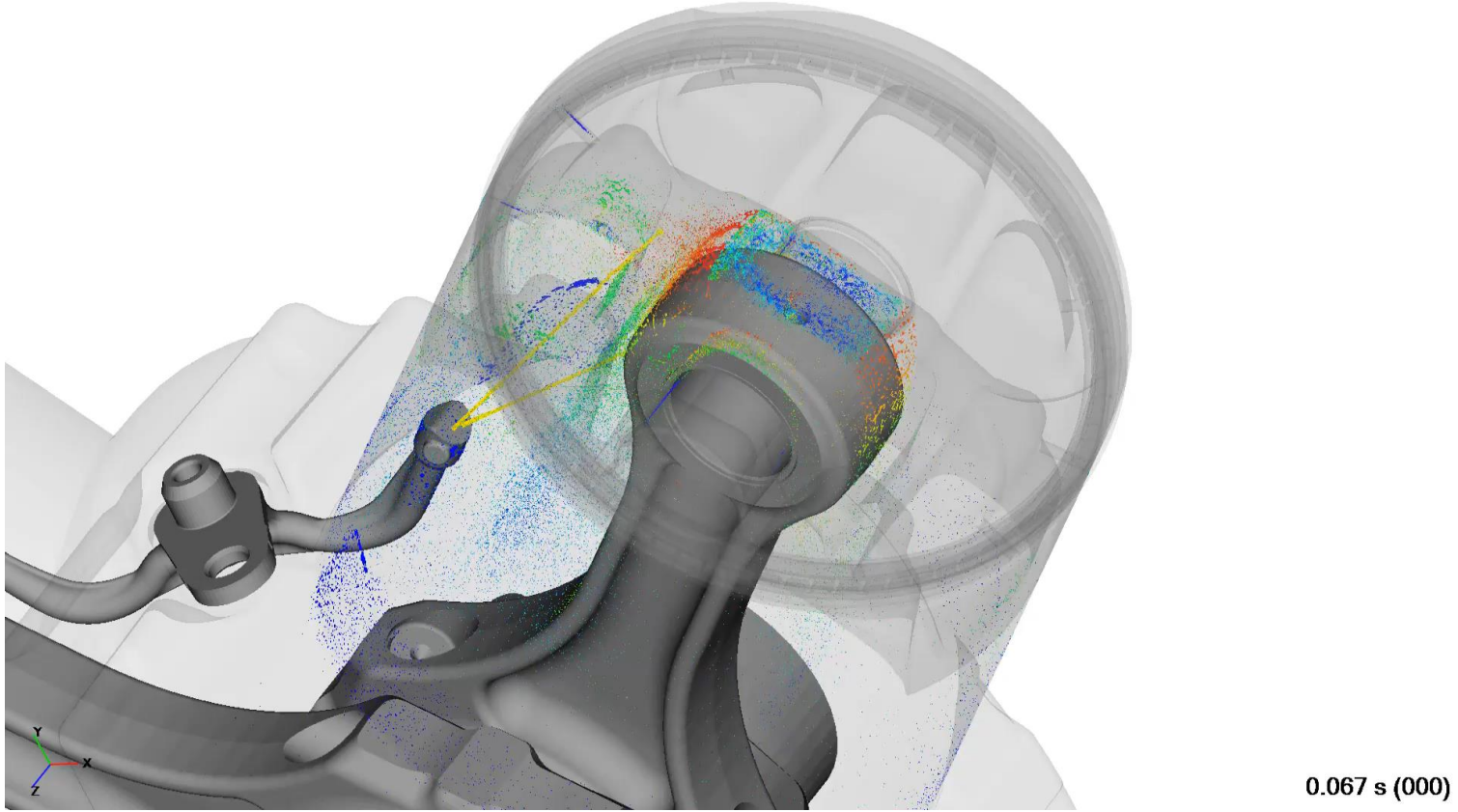
Results: proposal P1 (2X0.5)



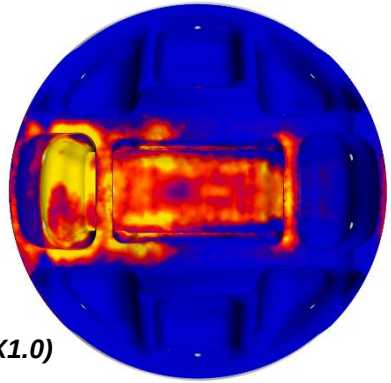
Results: proposal P2 (2X0.7)



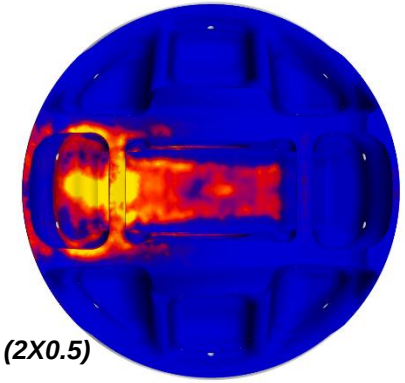
Results: proposal P3 (2X0.6)



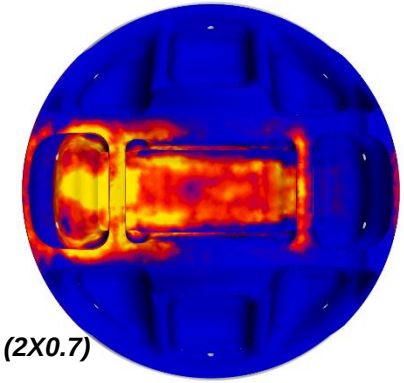
Maps of cycle average HTC @ 16000 rpm



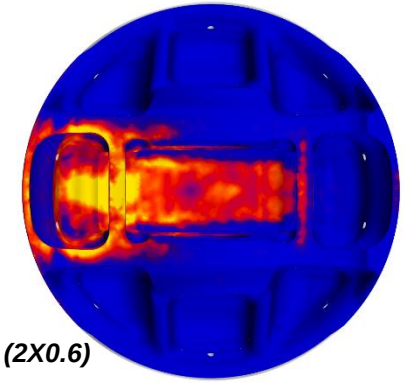
REFERENCE (1X1.0)



PROPOSAL P1 (2X0.5)



PROPOSAL P2 (2X0.7)

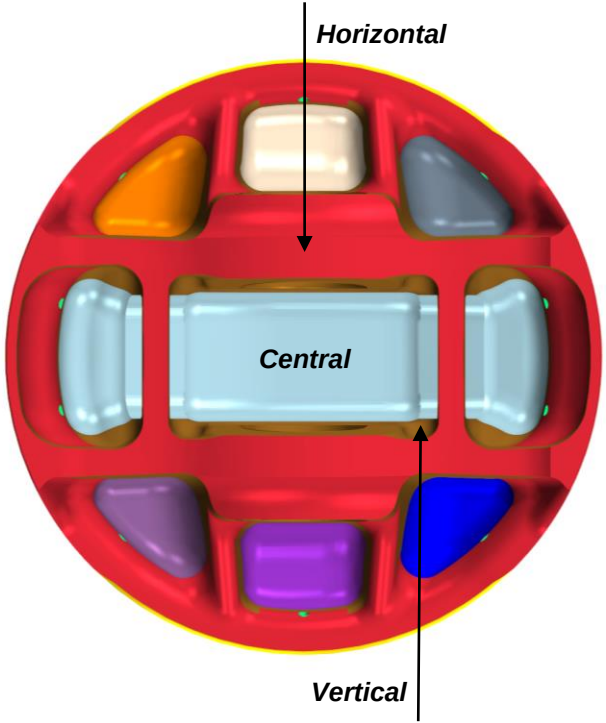


PROPOSAL P3 (2X0.6)

HTC quantitative comparison

- To compare the different layouts performances, it is convenient to extract a more quantitative measure: surface averaged heat transfer values were computed at the underside surface of the piston, appropriately divided into selected areas of interest.

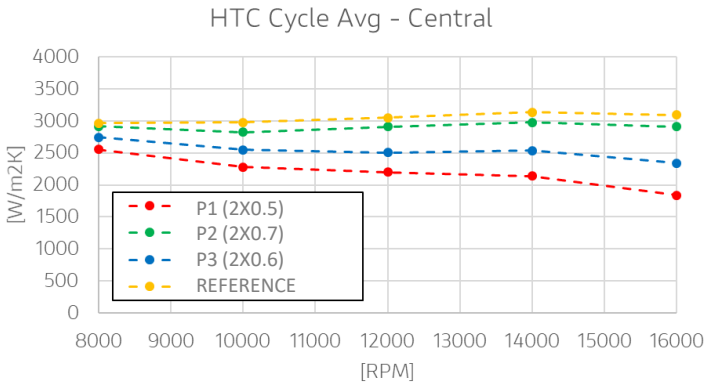
- Based on the maps shown in the previous slide, only the areas labeled as central (cyan), horizontal (perpendicular to cylinder axis, red) and vertical (parallel to cylinder axis, brown) in the picture on the right, are of interest. On the remaining surfaces the heat transfer is null.



Focus on Central Area

- The area previously called “central” is the surface where the heat transfer mainly takes place. This is due to the fact that the oil jets directly hit this area.

- As you can see in the graph, the reference layout had the highest coefficient values in this area. P2 proposal performed very similarly with an average difference of 4.5%. The remaining layouts showed more significant deviations in the flow rate trend, in particular the P1 proposal had the lowest flow rate.

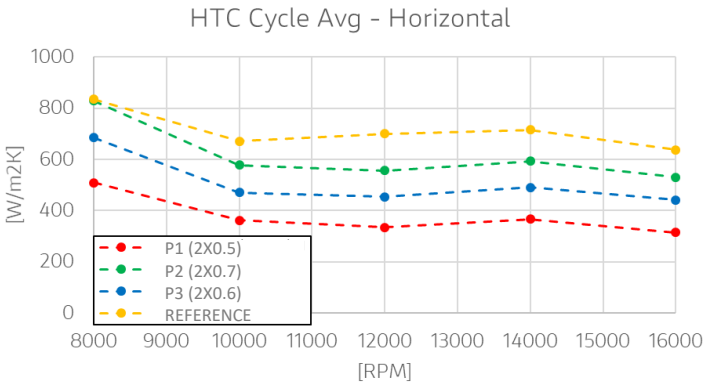


Focus on Horizontal Area

- Horizontal (and vertical) surfaces are not as important as the central area from a heat transfer point of view. Oil jets hit these surfaces for only a small time window of the engine cycle, this results in lower values of the heat transfer coefficient.

- Unlike the central area, the differences between the reference layout and the P2 proposal are more significant.

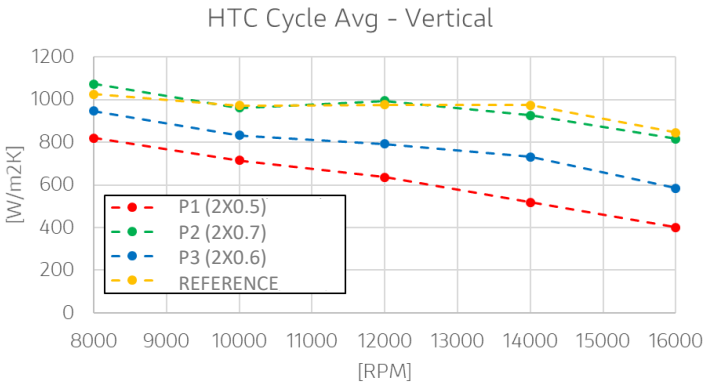
- In general, the trend of the curves seems to duplicate the flow rate one. This suggests that there is a close connection between the heat transfer coefficient and flow rate values for this area.



Focus on Vertical Area

- On the vertical surfaces the reference and P2 proposal curves are very similar. Having two holes seems to be beneficial in this area because it results in a more efficient distribution of the oil and, definitively, in more comparable heat transfer values. This despite of the flow rate gap.

- As already seen for the central and horizontal areas, heat transfer coefficient values of proposals P1 and P3 remain the lowest.



Conclusions

- Three different oil jet layouts have been studied with the main goal of identifying if one of them could reproduce the same heat transfer performance of a reference geometry but with lower oil flow rates. This taking advantage of the higher outflow velocities.

- However, for the layouts in question, the results have shown that the flow rate has a greater influence on the heat transfer coefficient values than velocities. Therefore, despite higher oil velocities, proposals P1 and P3 performed the worst.

- Layout P2 deserves a more detailed analysis. Its performance was comparable on the vertical area and slightly worst in the main central surface (-4.5% on average) with respect to the reference geometry. Assuming that the lower heat transfer coefficient values on the horizontal area are tolerable, the use of this layout would allow a reduction of the oil flow rate of about 20%. For this condition, the effects of a less efficient heat transfer on the integrity of the piston should be investigated.

Thank you for your attention